VCSEL Semiconductor PERFORMANCE and DEVELOPMENT (November 2025)

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Vertical-Cavity Surface Emitting Lasers (VCSELs) have proven to be effective in advancing high-speed optical communication and data transmission as well as various integrated photonic systems. Due to their extraordinarily small footprint, low power consumption, and easy scalability, they are perfect for these types of applications. This paper examines advances in VCSEL technology and how they can be implemented in other photonic systems. The fundamentals of the device are reviewed, focusing on the cavity structure and distributed Bragg reflectors (DBRs). Looking at other studies of this device showing that VCSELs can support speeds faster than 50 GB/s with great efficiency, simulations are explored regarding altering the distributed Bragg reflectors and cavity length to further improve performance. Results of the simulations show that optimizing the base structure of the VSCEL can reduce losses while maintaining a more pure beam structure. By improving DBR reflectivity and altering the cavity length, findings are promising in furthering development for the next generation of Vertical-Cavity Surface Emitting Lasers.

I. Device Fundamentals

THE Vertical-Cavity Surface Emitting Laser is a semiconductor laser diode. It emits a perpendicular laser beam (directly vertical instead of horizontal like an edge-emitting laser). It is further significant due to its extremely low current threshold current and pure Gaussian beam output. The architecture for the device is extremely small, allowing for easy integration in other photonic systems. Such applications include fiber optics, laser printers, 3D scans (LiDAR), and most commonly, the bottom of a computer mouse.

II. Structure and Function

The structure of a Vertical-Cavity Surface Emitting Laser is relatively similar to any semiconductor. There is a top and bottom contact, known as the ohmic contact, a top and bottom distributed Bragg reflector (a mirror), a p-type and n-type layer between those, and an active layer. The active region contains quantum wells, where the process of creating the laser beam happens. For VCSELs that have a wavelength range from 650nm to 1300nm, such materials used include Gallium Arsenide (GaAs) for the wafer, Aluminum or Gallium Arsenide for the DBRs, and (Au) for wire bonds. Current variance can be introduced with Aluminum Gallium Arsenide oxidation or using Ion Implants. The main benefit of a VCSEL's structure over a standard Edge-Emitting Laser is that it can be easily scaled at the wafer-level. An Edge-Emitting Laser's wafer needs to be precisely cut at an angle to produce a horizontal beam. But since a VCSEL is purely vertical, large arrays can be easily produced as its wafer needs one less level of processing.

A. Principles of Operation

Although the structure is different, a VCSEL and an Edge Emitting Laser produce a beam in a similar way. In the quantum wells, a process of recombination starts the chain reaction. Carriers, electrons and holes, are vertically injected into the active region through an electrical current. The forward bias of the devices across the p-n junction allows the electrons to flow from the n-type layer into the quantum wells. Similar to how the holes flow from the p-type layer into the valence band. When an electron combines with a hole in the quantum well, known as the process of recombination, it has a chance of spontaneous emission. The process in which that electron that combined with a hole can randomly release a photon. The next process that occurs in the cavity is stimulated emission. When an electron passes a photon, its energy level can drop down. In doing so, an identical photon to the one it passed, as in same energy level and direction of movement, is released. This is the most important process; it is essentially the chain reaction that allows for enough electrons to form a beam of light. Amplification is the last step where photons bounce off the DBRs in the active region creating more opportunities for stimulated emission. For the beam of light to actually escape the cavity, one of the DBRs (or mirrors) is slightly less reflective than the other. Allowing for a number of photons to shoot out of the cavity. For all this to occur, it is important that there is a population inversion. Meaning that the number of electrons in an excited energy state outnumbers the electron in a lower energy state. This is what prevents the beam creation process from collapsing.

B. Function

There are a number of use cases for this interesting device. Most notably, however, developments in VCSEL technology have primarily been led by the need for high-speed but energy-efficient data transmission. Traditional VCSELs have long been used in short-reach data links, however, the latest research extends the device's performance into multi-gigabit and even

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terabit communication systems. The IEEE study on high-speed VCSELs demonstrates devices achieving modulation speeds exceeding 50 Gb/s while maintaining extremely low power dissipation. A massive improvement that can be utilized in data centers and high-performance computing environments. These performance gains arise from optimized DBR mirror design, oxide aperture shaping, and thermal management strategies that reduce losses and improve modulation bandwidth.

VCSELs are not just seen in data transmission either. Work reported in the ResearchGate study expands the functional versatility of VCSELs beyond conventional means. These next-generation devices are being integrated into advanced photonic systems, such as on-chip optical networks, LiDAR emitters, and neuromorphic computer architectures. Their circular beam profile, wafer-level scalability, and low threshold current make them ideal for dense 3D integration and co-packaged optical systems.

Together, these innovations position the VCSEL not merely as a communication source, but also as core enablers of scalable, low-latency optical architectures, capable of bridging electronic and optical domains for next-generation data systems. The improvements explored in this report specifically enhanced DBR reflectivity and optimized cavity geometry, directly target further reduction in losses and improvements in bandwidth. Both of which are essential for sustaining high-speed and low-error optical data transmission.

III. Performance

The primary performance metrics used will come from the IEEE study regarding data transmission. It will connect to the simulations well.

A. *Equations*

Performance in Vertical-Cavity Surface-Emitting Lasers is typically evaluated in terms of modulation bandwidth, power efficiency, threshold current, output power, and thermals. Each influence the VCSEL's performance in high-speed data transmission. Zhang *et al.*'s IEEE study on energy-efficient VCSELs for optical interconnects reports small-signal modulation bandwidths exceeding 50 GHz, enabling data transmission rates above 100 Gb/s per channel. These results were achieved through optimized oxide aperture dimensions that minimize parasitic capacitance and series resistance, resulting in faster carrier dynamics and reduced RC time constants. Additionally, the devices demonstrate wall-plug efficiencies surpassing 40% and threshold currents below 1 mA, both of which are critical for reducing overall power consumption in large-scale optical networks. The ResearchGate study further highlights the importance of thermal management and reflectivity optimization for maintaining performance consistency across arrays and integrated systems. Increased DBR reflectivity and improved heat dissipation extend the stable operating range, reducing wavelength drift and maintaining single-mode operation under high current densities.

Collectively, these metrics underscore a VCSELs' potential as a compact and highly efficient device for short to long range data transmission. The simulation work below expands on these benchmarks by exploring how enhancing DBR reflectivity and tuning cavity length can further improve optical confinement and reduce threshold losses. Therefore, increasing modulation efficiency. Each of which being important factors when trying to sustain reliable terabit speeds in terms of communication performance.

IV. Simulations

A few simulations are run to determine if optimizing the structure of the VCSEL can improve its data transfer performance. A small signal modulated simulation is used with a bias current of 1.5mA and step current of 0.1mA, both practical values. The frequency response graphs are captured to determine the change in performance by observing the changes in bandwidth. First a VCSEL simulation is run with default parameters. Then only the bottom DBR reflectivity is raised to 99.99%. Next, the same increased reflectivity is kept but the cavity length is decreased to 100 x 10^{-7} cm from 900 x 10^{-7}cm. Next, the same increased reflectivity is kept again but the cavity length is increased to 900 x 10^{-5} cm from 900 x 10^{-7} cm. The difference in measurements is then compared.

Exp. 1 (Baseline)	Exp. 2	Exp. 3	Exp. 4
Parameters: Bottom DBR Reflectivity: 99.85% Cavity Length: 900x10^{-7} cm	Parameters: Bottom DBR Reflectivity: 99.99% Cavity Length: 900x10^{-7} cm	Parameters: Bottom DBR Reflectivity: 99.99% Cavity Length: 100x10^{-7} cm	Parameters: Bottom DBR Reflectivity: 99.99% Cavity Length: 900x10^{-5} cm
Figure 1	Figure 2	Figure 3	Figure 4

Only the bottom DBR reflectivity is increased because the top DBR still needs to remain less transparent to allow the beam to exit.

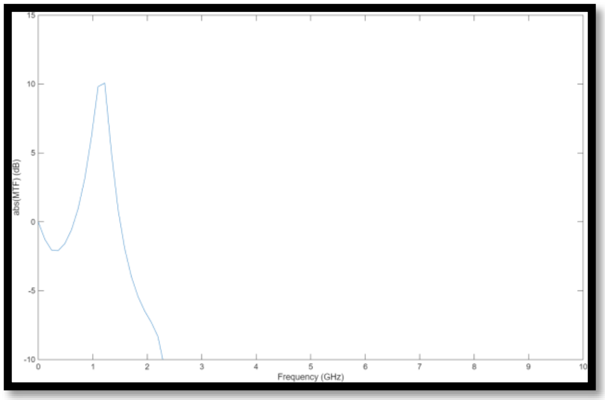

Figure 1: Exp. 1 → Default Parameters. Frequency Response. X-Axis = Frequency in GHz, Y-Axis = Magnitude of Modulation Transfer Function

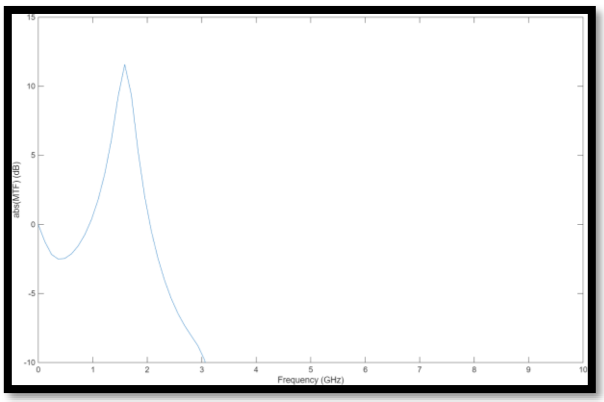

Figure 2: Exp. 2 → Increased DBR Reflectivity. Frequency Response. X-Axis = Frequency in GHz, Y-Axis = Magnitude of Modulation Transfer Function

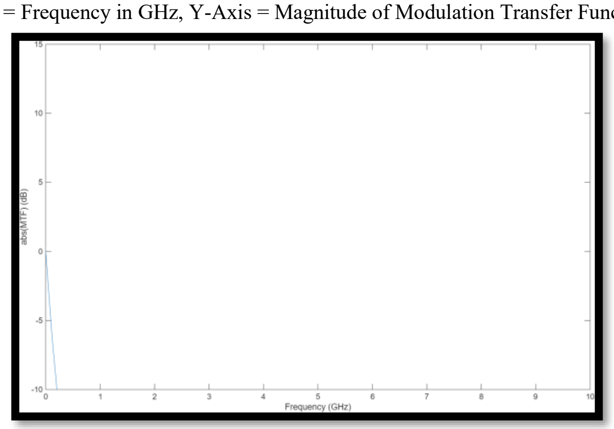

Figure 3: Exp. 3 → Increased DBR Reflectivity and Decreased Cavity Length. Frequency Response. X-Axis = Frequency in GHz, Y-Axis = Magnitude of Modulation Transfer Function

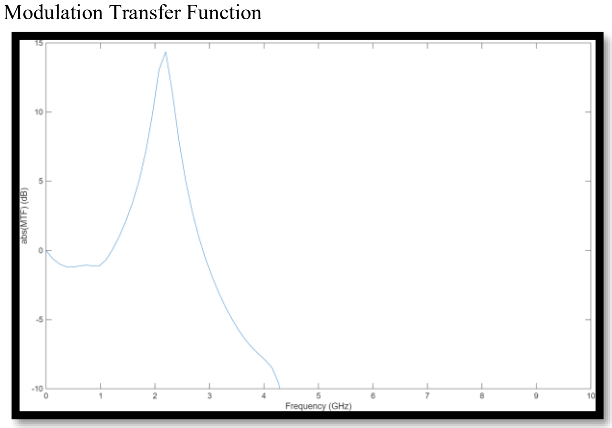

Figure 4: Exp. 4 → Increased DBR Reflectivity and Increased Cavity Length. Frequency Response. X-Axis = Frequency in GHz, Y-Axis = Magnitude of Modulation Transfer Function

V. Simulation Analysis and Conclusion

The default parameters set a baseline frequency of ~2.25 GHz. Increasing only the bottom DBR reflectivity reveals a large improvement increasing the device's bandwidth up to ~3.05 GHz. Keeping the improved performance by leaving the increased DBR reflectivity, the cavity length is then shortened. I expected that shortening the length would further increase bandwidth. But it instead causes the curve to completely collapse. However, increasing the cavity length instead revealed another substantial improvement in bandwidth as it appears to land near ~4.25Ghz. It also reveals a higher magnitude MTF (Modulation Transfer Function) peak of 15db over the previous 10-12db. This suggests there is less damping, allowing for improved modulation efficiency. However, a higher peak can also lead to unwanted oscillatory behavior and distortion. Although there may be imperfections in simulations, these results suggest that by optimizing the bottom DBR reflectivity while increasing the cavity length, enhancing photons confinement, the modulation bandwidth of the VCSEL doubles compared to the baseline.

If this optimized VCSEL structure were to be utilized in the high-speed modulation experiment, I expect a measurable performance increase. Further development and inclusion of this device can potentially lead to faster than terabit data transfer speeds.

This report explains the modulation characteristics of a VCSEL and how results were analyzed through simulations that explored the effects of DBR reflectivity and cavity length on optical performance. The default configuration exhibited a modulation bandwidth of approximately 2.25 GHz, serving as a baseline for comparison. After implementing an optimized VCSEL structure, the bandwidth increased to nearly 4.25 GHz, doubling the device's modulation bandwidth relative to its baseline. Since modulation bandwidth directly defines the upper limit of data transfer rate, these enhancements correspond to an increase in potential transmission speed from roughly 4 Gb/s to over 8 Gb/s (found from the equation below).

$$Maximum\ Data\ Rate \ \approx (1.5 - 2.0) \times f_{3dB}$$

And that improvement is for just one VCSEL. These results align with recent trends in newer studies where optimized DBR reflectivity and a refined cavity can allow for VCSELs capable of multi-gigabit data transmission. If this optimized structure were to be included in an array of devices like in Zhang *et al.*'s IEEE study, I believe petabit speeds could be within reach. Overall, the results demonstrate that structural tuning of the VCSEL can significantly increase performance and potentially offer a new path forward to faster and more efficient integrated photonic systems.

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